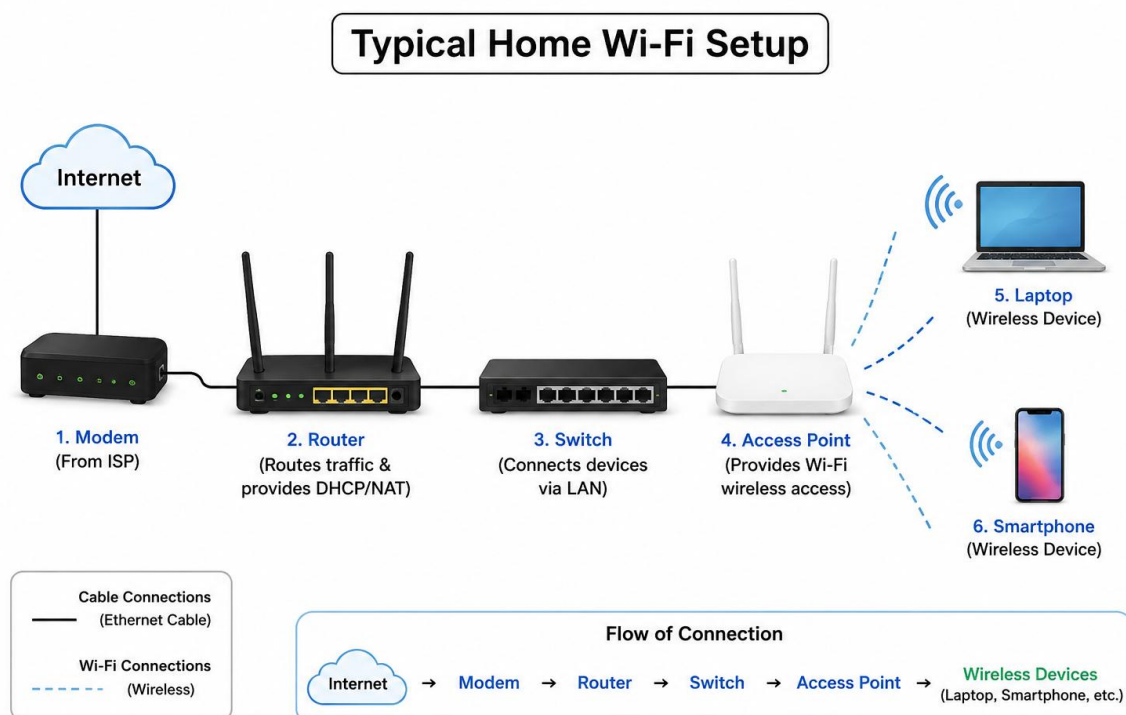


Assignment – 3.1

Q - 1. Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.



Q - 2. List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').

Device	Main Function	Real-Life Example
Modem	Connects your home to the Internet.	Used at home to get Internet.
Router	Shares Internet with many devices.	Used at home for phones and laptops.
Switch	Connects many wired devices together.	Used in an office to connect computers.
Access Point	Provides Wi-Fi to devices.	Used in schools and offices for Wi-Fi.
Network Cable	Connects devices and carries data.	Ethernet cable connecting a PC to a router.

Q - 3. Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.



Network Cable	Common Use
1. Ethernet Cable (Cat5e/Cat6)	Commonly used to connect computers, routers, switches, and other devices in a LAN/home network.
2. Coaxial Cable	Commonly used for cable TV, broadband internet, CCTV, and satellite connections.
3. Fiber Optic Cable	Commonly used for high-speed internet, data centers, and long-distance network connections.

Q - 4. Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.

Feature	Router	Switch
Main use	Connects different networks	Connects devices within the same network
Typical location	Home, office, or ISP network	Inside a LAN, office, or server room
How it handles data	Sends data between different networks using IP addresses	Sends data to the correct device using MAC addresses
Internet connection	Usually connects the local network to the Internet	Usually does not directly connect to the Internet
Devices connected	Connects networks and many devices	Connects computers, printers, servers, etc.
Example	Home Wi-Fi router sharing Internet	Office switch connecting 20 computers